

Professional Summary

Data Engineering Master's student at Constructor University with hands-on experience in Python, SQL, Apache Airflow, Apache Spark, Docker, PostgreSQL, ETL, and machine learning. Seeking opportunities in data engineering, data analytics, and related data-focused roles.

Technical Skills

Programming: Python, SQL, PHP, C++

Data Engineering & Processing: ETL/ELT, Apache Airflow, PySpark, Pandas, NumPy

Cloud & Infrastructure: Docker, Terraform, AWS

Version Control: GitHub

Professional Experience

Data Engineering Student / Projects *Constructor University* Sep 2024 - Present Bremen, Germany

- Developed data engineering and machine learning projects using Python, SQL, Airflow, Spark, and Docker.

IT Officer *KAKI Motors PLC* Nov 2023 - Aug 2024 Addis Ababa, Ethiopia

- Supported ERP systems and maintained IT infrastructure, user access, and system administration.
- Managed branch LAN networks and provided first-line hardware and software support.

IT Support Specialist *FH Ethiopia - International NGO* Nov 2021 - Oct 2023 Tigray, Ethiopia

- Provided technical support, system administration, and user-access management across departments.
- Managed participant data and trained data collectors on digital data-collection tools.

Software Developer *Allentech Electronic Technologies S.C.* Sep 2019 - Feb 2020 Tigray, Ethiopia

- Developed backend services and REST APIs for a digital ID card management platform using PHP/Lumen.
- Integrated application services and tested APIs using Postman in collaboration with frontend developers.

Selected Projects

Smart City Traffic ETL Pipeline *Python, Airflow, PySpark, Docker, PostgreSQL, Terraform, Superset*

- Designed and implemented a reproducible micro-batch ETL architecture for TomTom Traffic API data.
- Implemented Bronze/Silver/Gold processing, automated task retries, data validation, and multi-city workflows.

IoT Sensor Data Pipeline *Arduino, ESP32, Python, FastAPI, PostgreSQL, React*

- Built an end-to-end IoT pipeline to collect temperature and humidity data using Arduino/ESP32 and DHT11 sensors, and ingest it through a FastAPI REST API into PostgreSQL.
- Developed a React dashboard to visualize current and historical sensor readings, including minimum and maximum values.

Machine Failure Prediction *Python, Pandas, Scikit-learn, AI4I 2020 Dataset*

- Developed machine-learning models to predict equipment failure using the AI4I Predictive Maintenance Dataset.
- Applied data preprocessing, feature engineering, model training, and performance evaluation.

Gold Price ETL & Prediction Pipeline *Python, Apache Airflow, ETL, Sentiment Analysis, Machine Learning*

- Developed an automated Airflow ETL pipeline combining gold-price market data with real-time war/conflict news sentiment to predict whether gold prices will rise or fall on the next trading day.

Education

M.Sc. Data Engineering Sep 2024 - Present
Constructor University Bremen, Germany

Focus: Data processing, end-to-end pipelines, Machine Learning

B.Sc. Information Technology Sep 2015 - Dec 2020
Mekelle Institute of Technology / Mekelle University Tigray, Ethiopia

Languages

English - Fluent German - Beginner